Control-Anchored Assay Quality Monitoring

Biology / Anomaly Detection | Implementation and interview guide

Purpose and intended demonstration

Detect artificial assay outliers after correcting plate offsets with explicit reference controls. This is a computational biology learning project with synthetic or fictional inputs, not a clinical or experimentally validated system.

How the implementation works

Estimate each plate offset using designated controls. Learn a median and MAD-based feature scale from corrected training wells. Score a well by its maximum robust standardized deviation and choose a threshold from a separate normal validation plate.

Evaluation and evidence

Test on a held-out plate containing injected anomalies, keeping anomaly labels outside fitting and threshold selection. Compare with an always-normal baseline.

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Setup and commands

Requires Python 3.10 or newer and NumPy. Create a virtual environment: python -m venv .venv. On Windows use .venv\Scripts\python.exe if activation is unavailable. Install with python -m pip install -r requirements.txt. Run python app.py, then open results/report.html in your browser. Run python -m unittest discover -v to execute tests. The default seed is 42; change it with --seed. Replace sample_input.json using --input path-to-file.json, preserving its schema. Algorithm-only projects generate synthetic arrays from the seed rather than consuming external files.

Architecture and file responsibilities

CLI arguments -> JSON input or seeded generator -> domain pipeline -> validation or evaluation -> JSON and HTML artifacts. Imported functions can be used in a notebook or a separate application. Model checkpoints preserve learned arrays where applicable; see app.py for parameter order. This package does not include a production web service. Each project is independent and includes its own helper code so the ZIP can be extracted and run alone. app.py holds the experiment and domain functions: plate, correct, fit, scores, run. kit.py contains reporting, input loading and only the retrieval or generation helpers needed by this project. sample_input.json contains editable inputs. test_app.py checks domain behaviors; test_reporting.py checks metrics and HTML output boundaries. results/metrics.json records outputs and results/report.html provides a browser-readable report. Numeric projects also save arrays or learned parameters to results/artifacts.npz; inspect its keys with numpy.load. MODEL_CARD.md explains intended use and limits. .github/workflows/tests.yml runs tests and the fixture experiment. No API key is needed.

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Limitations and next improvements

The readouts and anomalies are mathematical fixtures. Control wells are assumed reliable; missing or contaminated controls invalidate the correction. Add multivariate covariance methods, control-failure checks, plate drift diagnostics and a licensed assay-QC benchmark.

Interview walkthrough

Present the problem and demonstrate one normal example, then one failure or boundary case. Explain the baseline or fixture mode, the split or validation design, and one limitation before discussing an extension. Be ready to trace data through the functions listed above. Explain why the chosen metric matches the task and why the available results cannot establish production performance. Add your own documented experiment and be precise about what you implemented or changed. These are AI-assisted educational projects; filenames and synthetic scores are not evidence of client work.

Research context and reading

These topics were selected from research directions reviewed on 22 October 2026. Recent activity suggests useful skills to develop; it is not a forecast of hiring, commercial success or scientific validity. The implementation is a small educational baseline and does not reproduce or claim the performance of any cited foundation model. Reading: Robust metabolomics normalization across experimental designs (2026): https://pubs.acs.org/doi/10.1021/acs.analchem.5c06841.

Verified build results

project

"Control-Anchored Assay Quality Monitoring"

test

{"accuracy": 1.0, "precision": 1.0, "recall": 1.0, "f1": 1.0, "tp": 12, "fp": 0, "fn": 0, "tn": 88}

always normal baseline

{"accuracy": 0.88, "precision": 0.0, "recall": 0.0, "f1": 0.0, "tp": 0, "fp": 0, "fn": 12, "tn": 88}

threshold

2.6696710042279124

validation normal wells

100

test wells

100

See results/metrics.json for all observations and TEST_RESULTS.txt for the executed tests. All datasets and replay responses shipped here are synthetic. Results can vary across seeds and numerical library versions.